Sarsengaliyev Damir

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Skills

Programming Languages: Python, SQL, Java, C++
Data & Analytics: SQL, Statistical Analysis, Hypothesis Testing (Kruskal-Wallis, Dunn, Granger Causality), Large-Scale Data Processing (DuckDB)
Machine Learning: Supervised & Unsupervised Learning, Clustering (K-Means, PCA/UMAP/t-SNE), Regression & Classification (XGBoost, LightGBM, CatBoost, Random Forest), Model Evaluation & Tuning
Frameworks & Libraries: Pandas, NumPy, SciPy, PyTorch, TensorFlow, Scikit-learn
Data & Experimentation: Dataset Curation and Preprocessing, Benchmarking, Experimental Design
Systems & Tools: REST APIs, Microservice-Based Systems, Git/GitHub
Languages: English (IELTS 7.5), Russian (Fluent), Kazakh (Fluent)

Technical Projects

Agentic Legal GraphRAG Pipeline June 2026

- Built an agentic GraphRAG system over 1,000+ Omani legal documents, a ReAct-style tool-calling agent (Groq function calling) that autonomously selects tools across hybrid retrieval, Neo4j graph traversal, and topic lookup to answer multi-hop legal questions (e.g. tracing amendment/repeal chains across laws)
- Hybrid retrieval: multilingual dense embeddings + BM25 sparse retrieval fused via Reciprocal Rank Fusion, with mode-aware cross-encoder reranking across Arabic and English indexes

Prediction Market Calibration Analysis April 2026

- Analyzed 679,643 Kalshi contracts and 72M trades to study how public attention affects prediction market calibration error using DuckDB for large-scale processing.
- Applied K-Means clustering with PCA, UMAP, and t-SNE to identify five distinct market taxonomies; validated via Kruskal-Wallis and Dunn post-hoc tests ($H=64{,}251$, $p=0$).
- Conducted Granger causality analysis between Google Trends scores and miscalibration; built FAISS similarity search index over 679K contracts for sub-millisecond calibration prediction

Household Income and Tenure Prediction (AHS 2023) April 2026

- Processed U.S. Census Bureau AHS 2023 dataset (48,527 units, 1,000+ variables) with structured leakage removal, missing value handling, and encoding across mixed feature types.
- Trained and compared five models (Linear/Logistic Regression, Random Forest, XGBoost, LightGBM, CatBoost); XGBoost achieved $R^2=0.334$ for income regression and AUC=0.974 for tenure classification.
- Conducted ablation study quantifying marginal value of person demographics; applied RandomizedSearchCV tuning reducing Random Forest overfitting gap from 0.61 to 0.08.

Activation Steering in Small LLMs October 2025 – December 2025

- Benchmarked GRPO-based LLMs on the MATH-500 and U-Math datasets, comparing DeepSeek-R1-Distill-Qwen-1.5B against a GRPO-trained Open-RS3 baseline.
- Identified skill-specific directions in the residual stream of Qwen3-4B, observing peak class separability in intermediate layers under non-thinking inference.
- Evaluated multiple activation steering strategies, including target–others mean, target–random mean, and correct–incorrect contrastive vectors.

Research Experience

Nazarbayev University Astana, Kazakhstan
Research Assistantship (with prof. Andrey Tkachenko) September 2025 - December 2025

- Preprocessed thousands of unlabeled audio survey responses for automatic transcription and timestamped segmentation using Whisper.
- Performed speaker diarization and segmented audio by speaker roles using Pyannote and WhisperX.
- Transcribed audio recordings with speaker labels, producing features for the preference-falsification analysis.
- Integrated a Qwen-2.5-3B LLM to clean and normalize noisy speech transcripts.

Nazarbayev University Astana, Kazakhstan
Research Assistantship (with prof. Adnan Yazici) September 2024 – May 2025

- Developed a multimodal emotion recognition model integrating video, audio, and text modalities using deep neural architectures (ConvNeXt for vision, RoBERTa for text, VGG16 for audio features).
- Achieved over 90% accuracy on the IEMOCAP benchmark, matching or surpassing state-of-the-art baselines.
- Deployed the trained model as a real-time video-based emotion classification system with a web interface.

Professional Experience

Freedom Holding Corp. Almaty, Kazakhstan
Data Scientist Aug 2026 – Present

- Built product matching, normalization, and clustering pipelines for categories, names, and attributes across millions of products; deployed to the production admin backend service.
- Developed ML models for service-adoption and churn prediction on a base of millions of users.

Smart Grades LLP Astana, Kazakhstan
Software Engineer December 2024 – July 2025

- Designed and maintained backend services supporting data-driven applications using Java Spring Boot.
- Built scalable REST APIs and microservices used for downstream analytics and model integration.
- Contributed to production-grade system reliability and data consistency in a distributed architecture.

Financial Monitoring Agency of the Republic of Kazakhstan Astana, Kazakhstan
Intern (Software Engineer) February 2024 – May 2024

- Developed automated data collection pipelines for digital asset investigations using Python.
- Extracted, cleaned, and structured large-scale web data for downstream analytical and investigative tasks.

Digital Economy Development Center LLP / Kazakhtelecom JSC Astana, Kazakhstan
Intern (Data Engineer) September 2023 – October 2023

- Engineered data processing pipelines handling millions of product records at a national scale.
- Optimized database queries and indexing strategies to improve data access latency.
- Automated computer-vision–based validation workflows for product labeling compliance.

Education

Nazarbayev University Astana, Kazakhstan
Bachelor of Science, Computer Science. GPA 3.61 / 4 Graduated June 2025
Relevant Coursework: Machine Learning: Theory and Practice, Deep Learning, Generative AI, Database Systems, Probability, Artificial Intelligence, Algorithms

References available upon request.